High-level maths questions list

Q	Main topic	Specifically	Marks
1	Differentiation	Implicit differentiation, chain and product rules, area under a curve	19
2	Differentiation	Differentiation from first principles, implicit differentiation, stationary points, second derivative, quotient rule	24
3	Trigonometry	Domain, trig manipulation, logarithms, differentiating trig functions, small-angle approximations	25
4	Algebra	Binomial approximation, integration by partial fractions	22
5	Algebra	Quadratic equations, arithmetic and geometric sequences, factor theorem, discriminant	28
6	Differentiation	Chain and product rules, simple trigonometry	28
7	Integration	Separation of variables, numerical integration	25

Question 1

a. Show that, for all valid x,

$$y = \sin^{-1} x \quad \Rightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}}$$

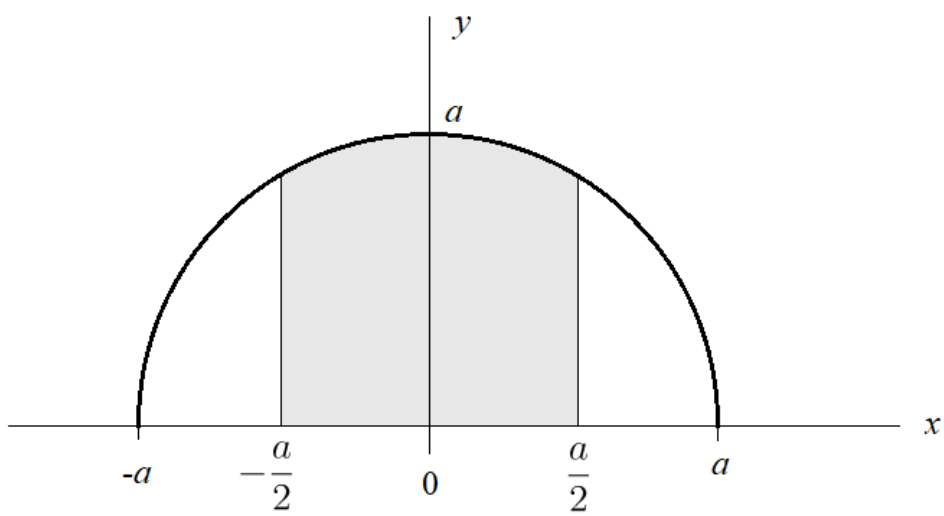
[5 marks]

b. For some constant a, let

$$f(x) = x\sqrt{a^2 - x^2} + a^2 \sin^{-1} \frac{x}{a}\ , \qquad |x| \leq a,\ a \in \mathbb{R}$$

i) Find the derivative of $f(x)$ with respect to x. [6 marks]

ii) The diagram shows a semicircle of radius a centred at the origin.



By considering your answer to part b.i), show that the area of the shaded region is $f(\frac{a}{2})$.

[4 marks]

iii) Find the proportion of the semicircle area which is shaded. Give your answer in exact form in terms of π and $\sqrt{3}$.

[4 marks]

[Total: 19 marks]

Question 2

a. Use differentiation from first principles to prove that

$$\frac{\mathrm{d}}{\mathrm{d}x}\cot x = -\csc^2 x$$

[7 marks]

b. The inverse of the cotangent function can be defined as

$$\cot^{-1} x = \tan^{-1}\frac{1}{x}\ ; \qquad x \in \mathbb{R}.$$

Use implicit differentiation to find the derivative of $\cot^{-1} x$ with respect to x.

[4 marks]

c. Show that the stationary points for the graph of $y = f(x)$ for

$$f(x) = \tan^{-1}\frac{1}{x^2} + \tan^{-1}\frac{1}{(x-a)^2}$$

have x-coordinates which satisfy the polynomial equation

$$2x^5 - 5ax^4 + 6a^2x^3 - 4a^3x^2 + (a^4+2)x - a = 0.$$

[5 marks]

d. When $|a| \leq \frac{2}{\sqrt[4]{3}}$, the graph of $y = f(x)$ has only **one** stationary point, which has x-coordinate $x = \frac{a}{2}$.

Determine the nature of this stationary point.

You should justify any case(s) when $f''(x) = 0$ with a calculation.

[8 marks]

[Total: 24 marks]

Question 3

For all real x, define the function

$$f(x) = \frac{1}{2} \ln \frac{1 + \sin x}{1 - \sin x}.$$

a. Consider the set S of all x such that $f(x)$ is undefined.

Write down all elements in the set $\{x \in S : |x| \leq 2\pi\}$. [4 marks]

b. By dividing the top and the bottom of the argument to the natural logarithm in $f(x)$ by a suitable trigonometric function, show that a simplified form of $f(x)$ is

$$f(x) = \ln |\sec x + \tan x|.$$

[8 marks]

c. Consider the function $g(x) = (1 - \sin x) f(x)$.

i) Using the form in part b), show that $f'(x) = \sec x$. [3 marks]

ii) Show that when x is small,

$$g'\left(x + \frac{\pi}{2}\right) \approx -\frac{1}{2}x + x \ln \left|\frac{2}{x}\right|.$$

[9 marks]

iii) Explain why the graph of $g(x)$ does **not** have a stationary point at $x = \frac{\pi}{2}$ despite its derivative approaching zero there.

[1 mark]

[Total: 25 marks]

Question 4

a. i) Find the first three terms, in ascending powers of *x*, of the series expansion of

$$\frac{\sqrt{1-x}}{\sqrt[3]{1+3x+x^2}}$$

[7 marks]

ii) Find the interval of validity for the approximation in part a.i).

[5 marks]

b. It is given that for all *x*,

$$\frac{\mathrm{d}}{\mathrm{d}x}\tan^{-1}x = \frac{1}{1+x^2}.$$

i) Show that

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,\mathrm{d}x = \frac{22}{7} - \pi$$

[8 marks]

ii) Explain how the result in part c) shows that $\pi < \frac{22}{7}$.

You should **not** evaluate the decimal forms of these values.

[2 marks]

[Total: 22 marks]

Question 5

It is given that there are four real solutions x to the equation

$$x^4 - 40x^2 + a = 0 \ , \qquad a \in \mathbb{R}.$$

a. Find the range of possible values of a. [6 marks]

b. When arranged from smallest to largest, the four roots are distinct and form an arithmetic sequence.

Find the value of a. [9 marks]

c. A student suggests there might also be a value of a such that the four roots instead form a geometric sequence with common ratio r (where $r \neq 0, r \neq 1$).

The student begins by letting the first term in the geometric sequence be A.

i) Explain why the order of the roots in the geometric sequence cannot be from smallest to largest as in part b).

[1 mark]

ii) By considering separately the product and sum of the four roots and applying the factor theorem to the equation $x^4 - 40x^2 + a = 0$, show that

$$a = r^6 A^4 \quad \text{and} \quad r = -1.$$

[7 marks]

iii) Hence show that the only case in which the roots form a geometric series is when a = 400.

[4 marks]

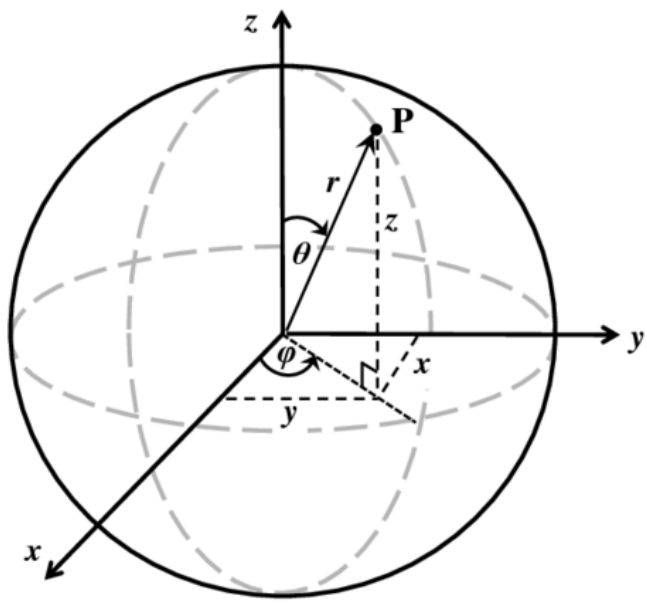
iv) Comment on another way the four roots are related when a = 400.

[1 mark]

[Total: 28 marks]

Question 6

The point *P* lies on the surface of a sphere of radius *r* centered at the origin of a 3 -dimensional (x, y, z) Cartesian coordinate system. The angles θ and ϕ are defined relative to the positive axes as shown in the diagram below.



a. Considering the triangles shown with dashed lines above, find expressions for

i) r [1 mark]

ii) $\cos\theta$ [2 marks]

iii) $\tan\phi$ [1 mark]

in terms of x, y and z.

b. Hence, or otherwise, find expressions for

i) x [2 marks]

ii) y [1 mark]

iii) z [1 mark]

in terms of r, θ and ϕ.

c. A differential equation representing the variation of a physical quantity, u, as a function of the position (r, θ, ϕ) in space, is

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}u}{\mathrm{d}r}\right) + \frac{1}{r^2\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}u}{\mathrm{d}\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\mathrm{d}^2u}{\mathrm{d}\phi^2} = 0$$

(Equation 1)

You may assume that r, θ and ϕ can vary independently, and therefore when differentiating u with respect to any of these variables, **all other variables should be considered constant**.

Verify that

$$u = \left(r + \frac{1}{r^2}\right)\sin\theta\cos\phi$$

is a solution to *Equation 1*.

[10 marks]

d. Using the substitutions found in part a), write the solution for u in part c) in terms of x, y and z and hence verify that it satisfies the differential equation

$$\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + \frac{\mathrm{d}^2u}{\mathrm{d}y^2} + \frac{\mathrm{d}^2u}{\mathrm{d}z^2} = 0$$

[10 marks]

[Total: 28 marks]

Question 7

A pendulum consists of a mass attached to one end of a light inextensible string of length L. The other end is fixed to a frictionless pivot. The mass is lifted through an initial angle of θ_0 to the vertical. It is released from rest at time t = 0.

The subsequent motion of the pendulum is described by the differential equation

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \frac{g}{L}\sin\theta = 0 \qquad \textit{(Equation 2)}$$

where θ is the angle between the string and the vertical. This model does **not** assume that θ_0 is small, but does assume the motion of the pendulum is periodic (neglects air resistance and other damping effects.)

a. By multiplying both sides of *Equation 2* by $\frac{d\theta}{dt}$, show that the value of

$$\frac{1}{2}\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 - \frac{g}{L}\cos\theta$$

is constant throughout the motion of the pendulum. [4 marks]

b. Hence, by considering the initial conditions of the pendulum, show that the first-order differential equation modelling the motion for all t > 0 is

$$\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 = \frac{2g}{L}\left(\cos\theta - \cos\theta_0\right) \qquad \textit{(Equation 2)}$$

[2 marks]

c. The time period of the pendulum can be calculated using *Equation 2* (above) and the modelling assumptions given.

Starting from *Equation 2*, use separation of variables followed by a suitable trigonometric identity and the substitution

$$\sin\frac{\theta}{2} = \sin\frac{\theta_0}{2}\sin x$$

to show that the time period for one full oscillation, *T*, of the pendulum is given by

$$T = 4\sqrt{\frac{L}{g}} \times \int_0^{\frac{1}{2}\pi} \frac{1}{\sqrt{1-\sin^2\left(\frac{1}{2}\theta_0\right)\sin^2 x}}\,\mathrm{d}x$$

.

[12 marks]

You are advised to plan your approach before beginning. You are **not** required to justify rejecting negative solutions when square-rooting *Equation 2*.

d. A particular pendulum has a length of 0.500 m and is released from an initial angle of 60° to the vertical.

The equation for the time period from part c) has been repeated here to help you.

$$T = 4\sqrt{\frac{L}{g}} \times \int_0^{\frac{1}{2}\pi} \frac{1}{\sqrt{1-\sin^2\left(\frac{1}{2}\theta_0\right)\sin^2 x}}\,\mathrm{d}x$$

i) Using the trapezium rule, estimate the time period of this pendulum.

You should use 4 strips of equal width and take g = 9.81 ms^{-2}. Give your answer to the greatest possible degree of precision.

[4 marks]

ii) Using a calculator with a numerical integration function, write down the exact value of the time period. Comment on your result and suggest a reason for the accuracy of your answer.

[3 marks]

[Total: 25 marks]